Article

Luminos : a Doctor Strange Inspired Device

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Abstract. This project presents the design and implementation of an infrared (IR)-based LED control system that enables wireless control of lighting patterns using gesture commands. The objective of the study is to develop a simple and low-cost communication method that allows commands to be transmitted from a wearable controller to multiple LED lantern units using infrared transmitters and receivers. The system integrates an ESP32 microcontroller, an MPU6050 motion sensor, infrared communication modules, and NeoPixel LED rings.

The wearable control device detects hand gestures using the MPU6050 sensor by analyzing acceleration data, particularly along the x-axis and the overall magnitude of motion. Detected gestures are translated into infrared commands using the NEC communication protocol and transmitted through an IR emitter. Each Smart LED Lantern is equipped with an ESP32 microcontroller and four infrared receivers to ensure reliable signal detection from different directions. When an IR signal is received, the ESP32 decodes the command and triggers the corresponding lighting response on the NeoPixel LED ring.

Experimental results show that the system successfully converts hand gestures into wireless control commands with stable communication and low response latency. The LED lanterns respond reliably to commands for color change, white light intensity adjustment, and on/off toggling. The results demonstrate that the proposed system provides an intuitive and interactive method for lighting control while maintaining low hardware complexity and cost, making it suitable for smart lighting and human–machine interaction applications.

Keywords: infrared communication, gesture recognition, ESP32 microcontroller, embedded systems, NeoPixel LED ring, smart lighting control

1. Introduction

Infrared Ray is widely used for short wireless transmission. Home appliances, such as remote controls, use this technology due to its low cost and simple implementation. Infrared Rays are used in this project to send commands from Hand Device to Smart LED Lanterns to turn on/off and adjust the light colors.

The project consists of one Hand Device, with one IR sender, and three Smart LED Lanterns, consisting of four IR receivers. Both Neopixel LED and IR receiver are connected to ESP32, which acts as a core facilitator of the Smart LED Lantern functionality. Once the Smart LED Lantern detects incoming Infrared Rays from the Hand Device, the IR receiver sends the signal to the ESP32. Then, the signal from the IR receiver is forwarded to NeoPixel LED to either turn on/off or adjust its color via ESP32.

2. Literature Review

Infrared communication functions by transmitting light signals in the infrared spectrum. The signals are modulated at a specific frequency of 38kHz. It allows the receiver to differentiate the signal from ambient light sources.

The IR receiver modules consist of photodiodes and signal-processing circuits that filter and detect modulated infrared signals. The module generates microcontroller-readable signals when a valid signal is detected.

The microcontroller used in this project is ESP32. This type of microcontroller is widely used in embedded system applications and projections because of the flexible input/output interfaces and the processing capability. Signals from sensors, process data, and control various electronic components can be read by the ESP32.

The addressable LED rings used in this project is Neopixel LED ring 16. One Neopixel LED ring consists of 16 individual LED lights. Each LED light can be controlled through a single data line and display different colors and brightness levels, allowing complex lighting effects and animations.

Additionally, the gyroscope sensor used in this project is MPU6050. It is a 3-axis accelerometer and a 3-axis gyroscope. This MPU6050 is able to read acceleration data and angular velocity across 3 axes(x y z) and communicates with the ESP32 through the I²C interface.

Therefore, by integrating IR receivers, microcontrollers, addressable LED rings, and MPU6050. It is possible to create interactive systems that respond dynamically to wireless commands.

3. System Design

3.1. System Architecture

The system consists of five main components: infrared receivers, infrared emitter, microcontrollers (EPS32), a gyroscope (MPU6050), and LED rings (Neopixel Led ring 16).

First of all, the MPU6050 detects the hand gestures to control the light bulb. Since the remote only controls the motion by measuring the arm rotation and movement, the system mainly analyzes the acceleration along the x-axis to determine them. Therefore, if the acceleration exceeds a certain level, the mpu6050 will send the corresponding control commands through the emitter.

Additionally, the MPU6050 detects the magnitude of the total acceleration and its change over time to measure the sudden motion. Thus, when a significant motion is detected, a particular command is sent. The identified gestures are transmitted using an infrared (IR) transmitter with the NEC protocol.

Subsequently, the IR emitter transmits the infrared signals to the IR receivers. Each IR receiver is connected to a digital input pin of the microcontroller. In this project, pins D15 are connected to the IR receivers for each microcontroller. When an IR signal is transmitted, the microcontrollers decode the signal using the IRremote library.

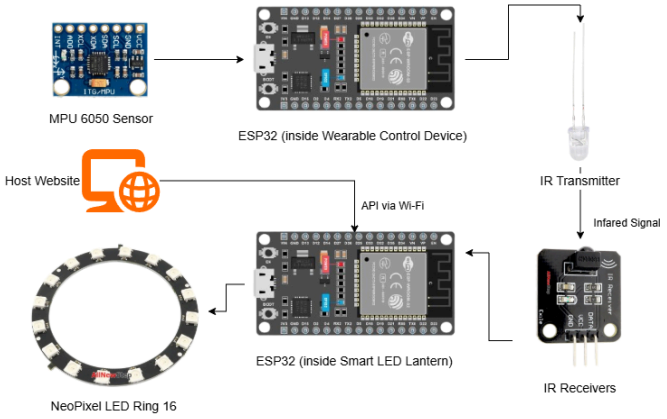
After the decoding process, the program identifies the meaning of the command and triggers the corresponding LED feature. The LED rings used in this project consist of multiple individually addressable LEDs. The results and signals are explained in Table 1.

Table 1. Mapping between detected gestures, IR commands, and LED responses

Motion	IR Command	Resulting LED Feature
Neutral	None	LED status remains unchanged
Tilt Right	1	Change color
Tilt Left Left	2	Change white level
Sudden motion	3	Toggle ON/OFF

The system software consistently tracks the IR receivers for incoming signals. Once the signal is detected,the LED features are executed by the program, including color changes, toggle on/off, and white level change. This design allows the LED system to respond instantly to IR commands.

Fig. 1. Block diagram of IR-WiFi LED control system



3.2. Wearable Control Device Design

3.2.1. Electronic Hardware Design

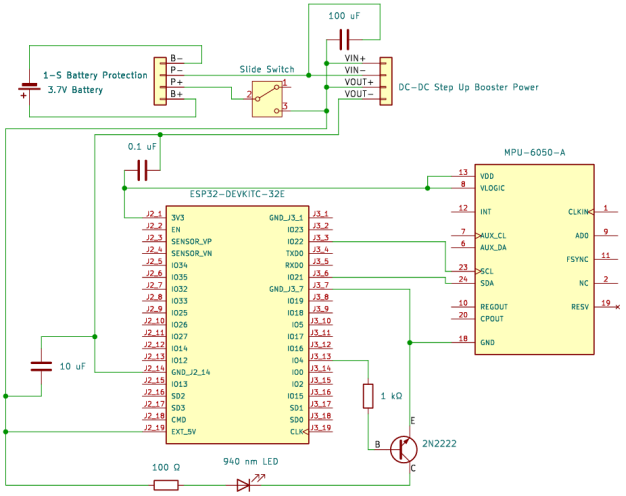
The electronic hardware of the wearable part consists of an ESP32 microcontroller, an MPU6050 motion sensor, an infrared transmitter, a slide switch, and a power supply module.

The power supply module consists of a 1000 mAh 3.7 V lithium battery, a 1-series lithium battery protection module, and a 2A DC-DC step-up booster power module (MT3608). The main function of the 1-series lithium battery protection module is to protect the battery from overcharging, over-discharging, short circuits, and overcurrent, while the 2A DC-DC step-up booster power module increases the voltage from 3.7 V to 5 V, which can be supplied to the Vin pin of the ESP32.

The 940 nm IR LED, which serves as the infrared transmitter, is driven by a 2N2222 NPN transistor. The transistor acts as a switching amplifier controlled by the ESP32 to provide sufficient current to the IR LED for reliable infrared signal transmission. A current-limiting resistor and a capacitor are used to protect the LED and ensure stable operation.

The gyroscope (MPU-6050) is connected to the ESP32 through SDA (Serial Data Line) and SCL (Serial Clock Line) to read acceleration and angular velocity data for gesture detection.

Fig. 2. Circuit schematic of wearable control device



3.2.2. Mechanical and CAD Design

The wearable control device consists of 2 parts: the enclosure and the base. The enclosure is designed to house the main electronic components of the wearable controller, including the ESP32 microcontroller, MPU6050 motion sensor, and infrared transmitter circuit. Mounting posts with screw holes are integrated into the enclosure to allow secure fastening of the circuit boards and structural components. A circular opening and mounting plate are provided to accommodate the infrared transmitter assembly, ensuring proper alignment for signal transmission.

Fig. 3. Three-view CAD design of the wearable control device enclosure

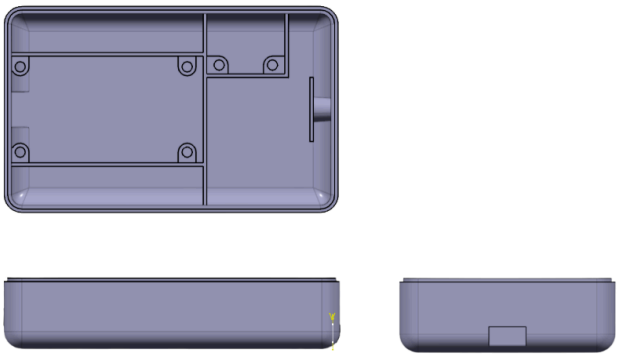


Fig. 4. Internal compartment design of the wearable control device enclosure



On the other hand, the battery module and power management components are placed inside the base. Rectangular slots are incorporated on the floor of the base to allow straps to pass through the housing.

Fig. 5. Three-view CAD design of the wearable control device base

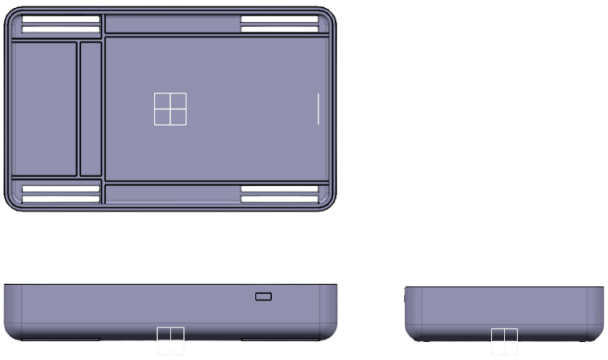


Fig. 6. Internal compartment design of the wearable control device base



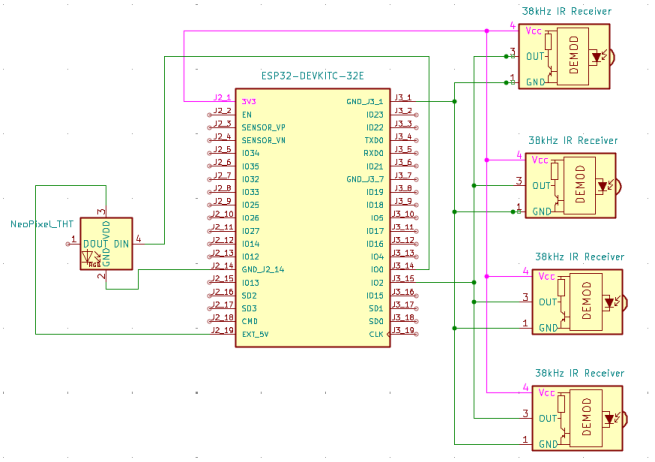
3.3. Smart LED Lantern Design

3.3.1. Electronic Hardware Design

The electronic hardware of the lantern part includes ESP32 microcontroller and 4 pieces of infrared receivers, and NeoPixel LED ring.

The receiver system uses multiple infrared receiver modules to detect the transmitted IR signals. Each receiver is connected to a digital input pin of the ESP32 microcontroller.

Fig. 7. Circuit schematic of Smart LED Lantern



3.3.2. Mechanical and CAD Design

The smart LED lantern comprises 2 parts: base and head. Both parts are connected by thread. Inside the base, a rectangular opening on the side wall of the enclosure is designed to expose the infrared (IR) receiver sensor to the external environment. This opening allows incoming infrared signals from the wearable transmitter to reach the receiver without obstruction, ensuring reliable signal detection. A cantilever support structure is

integrated to hold the IR receiver module in a fixed position.

Fig. 8. Three-view CAD design of the Smart LED Lantern base

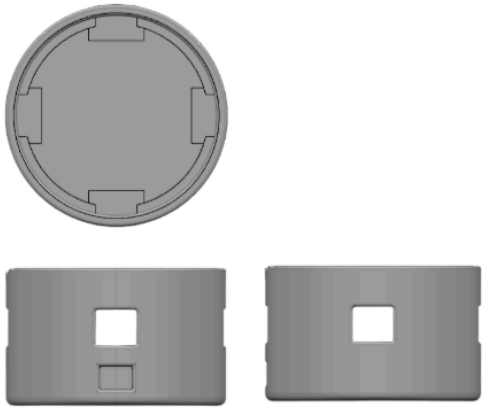


Fig. 9. Internal cross-sectional view of the Smart LED Lantern base

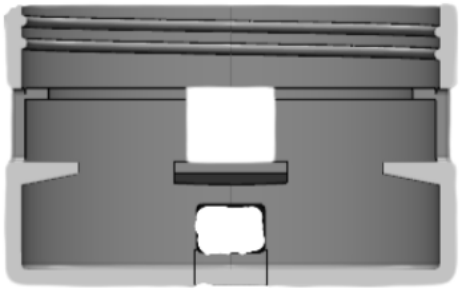


Fig. 10. Two-view CAD design of the Smart LED Lantern base

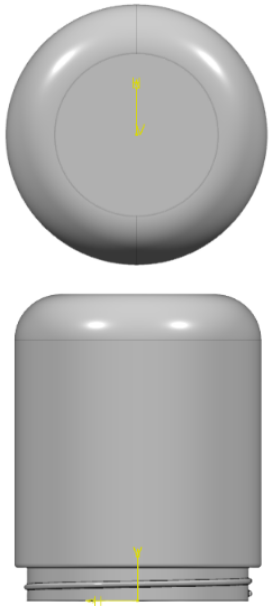
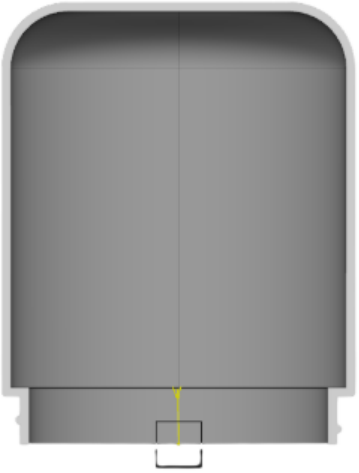


Fig. 11. Internal cross-sectional view of the Smart LED Lantern head



4. Results

4.1. Gesture Detection and Signal Transmission

The implementation of the IR-based LED control system using the ESP32 microcontroller, MPU6050 gyroscope, and NeoPixel LED strip yielded the following functional results. The system successfully translated physical hand gestures into wireless commands and visual lighting effects.

The MPU6050 sensor accurately captured motion data, specifically focusing on the x-axis acceleration and total magnitude of movement. The system successfully categorized these movements into three distinct IR commands using the NEC protocol:
Tilt Orientation: Hand rotations (Tilt Right and Tilt Left) were reliably converted into IR Commands '1' and '2'.
Dynamic Motion: Sudden movements were detected by monitoring the change in total acceleration over time, triggering IR Command '3'.
The infrared emitter maintained a stable connection with the receivers, ensuring that the modulated 38 signals were transmitted with minimal latency.

4.2. IR Decoding and System Response

The four IR receivers (all IR are connected to pins D15) successfully captured the transmitted signals. The IRremote library effectively decoded the incoming code of 1 2 and 3, allowing the microcontroller to map each command to its designated LED function.

4.3. LED Visual Output

The NeoPixel LED rings demonstrated high responsiveness to the decoded commands.

Upon receiving Command 1, the addressable LEDs changed colors instantly, demonstrating the microcontroller's ability to update the data line of the NeoPixel strip in real-time.

Command 2 allowed for granular control over the white light intensity, proving the system's capability to handle PWM-like (Pulse Width Modulation) effects through IR triggering.

The toggle function (Command 3) operated without debouncing issues, providing a clean "On/Off" state transition even during rapid physical movements.

4.4. Portability of the Hand Device

Multiple electrical components were used and adjusted for the Hand Device. With adjustments and elimination of unnecessary components, the removal of NPN transistors, the device achieved the compact dimension attached to the user's arm. The 3.7V 1000mAh lithium battery is also used to power the ESP32 sufficiently with the needs while ensuring space efficiency.

4.5. Circuit Integration of Hand Device

The power source and the microprocessor are connected within the hand device with seamless disruptions. However, prior to that, the step-up booster connected with the battery and the ESP-32 generated an unexpected electromagnetic interference, which was identified and later removed. The lithium battery is then connected directly with the Vin of the ESP-32.

The integration of the MPU6050 and the IR communication protocol created a seamless interface. The system proved that embedded sensors could effectively replace traditional remote buttons, providing a more intuitive and interactive way to manage ambient lighting. The use of multiple IR receivers ensured high signal coverage, allowing the LED rings to receive commands from various angles.

5. Discussions

The integration of the MPU6050 with IR communication demonstrated high responsiveness, though it required precise calibration to manage sensor sensitivity. Since the MPU6050 is prone to noise, we implemented a specific acceleration threshold to distinguish between simple "Tilt" commands and deliberate "Sudden Motion." This ensured that lighting toggles only occurred during intentional movements.

To overcome the inherent "Line-of-Sight" limitations of infrared technology, a quad-receiver array was utilized. This expanded the system's field-of-view, allowing for reliable signal decoding even without perfect transmitter alignment. Furthermore, while the system achieved a near-instantaneous latency of under 100ms, future optimizations could focus on power management. By transitioning from constant signal polling to interrupt-based sleep modes, the ESP32's energy efficiency could be significantly improved for long-term applications.

During initial implementation, the step-up booster was used to regulate the voltage to 5V before supplying it to the ESP32. However, when both circuits are integrated, the system becomes inoperable. Further investigation points out that the electromagnetic wave generated by the step-up booster interferes with the electrical circuits. The step-up booster is later removed, and the battery is connected directly to the Vin pin of the ESP32.

6. Conclusion

This project successfully demonstrated the development of an IR-based gesture-controlled lighting system. By combining an MPU6050 gyroscope, ESP32 microcontrollers, and addressable NeoPixel LED rings, we created a functional prototype that translates physical motion into wireless lighting commands.

For software, the system successfully identified three distinct gestures (Tilt Right, Tilt Left, and Sudden Motion) and mapped them to specific LED behaviors (Color Change, White Level, and Power Toggle). The use of the NEC protocol and multiple IR receivers proved to be a reliable method for short-range wireless communication in an embedded environment. The modular nature of the code allows for easy expansion. More gestures or complex lighting patterns such as animations or rainbow cycles could be added without changing the hardware architecture.

On the other hand, the hardware adjustments, including pin assignments and the selection of compact electronic modules, resulted in space efficient product design.

6.1. Suggestion for Software development

The MPU6050 sensor produces readings that can contain noise and small unintended movements. These fluctuations can affect gesture detection accuracy. Future development can apply digital filtering techniques to improve the stability of the sensor data. Examples include moving average filters and low-pass filters. These filters smooth the acceleration values before gesture classification. This process reduces false detections and improves overall system stability.

The current system mainly uses acceleration along the x-axis to detect gestures. The system also relies on simple threshold values to classify motion. Future versions of the software can analyze acceleration from the x, y, and z axes together. Multi-axis analysis can support more reliable gesture recognition. This approach can also enable additional gestures. Examples include circular motion, double shake, and directional sweeps.

The current program continuously polls the IR receivers to detect signals. This method requires the

microcontroller to check the receivers repeatedly. Future versions of the system can use interrupt-based detection. In this approach, the microcontroller responds only when an interrupt event occurs. This method reduces processor workload. It can also improve power efficiency, especially in battery-powered systems.

Infrared communication can sometimes be affected by noise or unintended signals. The system can improve reliability by adding command verification routines. For example, the software can check repeated commands before execution. The system can also apply checksum validation to verify signal integrity. These techniques help prevent false triggering.

The current NeoPixel LED control system supports basic lighting behavior. Future software development can add more advanced lighting effects. These effects can include:

- fading transitions
- rainbow animations
- brightness gradients

The program can organize each lighting effect into separate functions. This structure makes the code easier to maintain and extend.

The program structure can benefit from a more modular design. The system can separate different functions into independent modules. These modules can include sensor processing, IR communication, LED control, and gesture recognition. A modular design improves code readability. It also simplifies debugging and testing. This structure allows developers to add new features without modifying the entire codebase.

6.2. Suggestion for Hardware development

Although the use of four infrared (IR) receivers improved the signal reception angle, the system could be further enhanced by increasing the transmission strength of the IR emitter. This could be achieved by using higher-power IR LEDs or implementing a transistor driver circuit to supply higher current to the emitter, which would enable more reliable communication over longer distances and reduce sensitivity to the orientation of the wearable controller.

To improve usability and compactness, the handheld device could also be redesigned using a custom printed circuit board (PCB) instead of multiple module boards and jumper wires. Integrating the ESP32, MPU6050 sensor, and IR transmitter circuitry onto a single PCB would reduce the overall device size, improve electrical reliability, and make the wearable device more comfortable for users.

Furthermore, the lighting system could be expanded beyond portable lanterns to other types of lighting devices such as ceiling lights or wall-mounted lighting systems, allowing more complex lighting patterns and broader applications. The system could also be extended to interactive uses, for example by creating a simple game where multiple lanterns randomly turn on and players use the wearable controller to turn them off within a limited time, with the player who switches off the most lights being the winner.

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